

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

Nov-Dec. 2013

EE102: FUNDAMENTALS OF ELECTRICAL ENGINEERING

Date: 05/12/2013

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the Kirchhoff's Current law and Voltage Law. [03]
- (b) Explain the methods of transforming a star network of resistances into delta network and vice versa and derive necessary expressions. [04]
- Q - 2 (a) Compare Electric and Magnetic Circuit with similarities and dissimilarities. [07]
- (b) With Reference to electrostatic and capacitance: [07]
- (i) State Coulomb's Laws
 - (ii) Define: Electric field Intensity, Electric potential, potential Gradient, Permittivity, and Capacitance

OR

- Q - 2 (a) Explain the charging and discharging of a capacitor, C, through a resistor R, with neat sketch and derive the equation $V_c = V (1 - e^{-t/RC})$. Assume that the R-C series circuit is connected across a d.c. supply of voltage V. [07]
- (b) With Relevant figures, Explain in brief: (i) Self Inductance (ii) Mutual Inductance. [07]
- Q - 3 (a) A ring has a mean diameter of 21 Cm and a cross sectional area of 10 cm^2 . The ring is made up of semi circular sections of cast iron and cast steel, with joint having reluctance equal to an air gap of 0.2mm. Find the amp-turns required to produce a flux of $8 \times 10^{-4} \text{ Wb}$. The relative permeability of cast steel and cast iron are 800 and 166 respectively. Neglect fringing and leakage effect. [07]
- (b) Coil A and B with 50 and 500 Turns respectively are wound side by side on a closed iron circuit of section 50 cm^2 and mean length of 1.2m. Estimate (i) Self Inductance of each coil, (ii) Mutual Inductance between coils, (iii) emf induced in coil A if the current in coil B grows steadily from 0 to 5 Amp in 0.01 Sec. Assume relative permeability of iron as 1000. [07]

OR

- Q - 3 (a) What is Capacitor? Derive the expression for the equivalent capacitance of Capacitors connected in Parallel and in series. [07]

- (b) State and Explain the Faraday's Laws of Electromagnetic Induction and also explain Fleming's Left Hand Rule and Right hand rule in brief. [07]

SECTION – II

Q - 4 (a) Give the comparisons of A.C. and D.C. [04]

(b) Define the following terms [03]

I. Cycle

II. Amplitude

III. Instantaneous

Q - 5 (a) Find out the average value of an alternating quantity with Mid-ordinate method. [07]

(b) Define the form factor and peak factor [07]

OR

Q - 5 (a) Derive expressions for impedance, power factor & current for an R-L series circuit when an A.C voltage is applied to it. Draw also the phasor diagram. [07]

(b) Explain resonance in series RLC circuit & its Q-factor. [07]

Q - 6 (a) Explain the term power factor in connection with A.C circuit [07]

(b) Define the voltage and current relationship in star connection three phase [07]

OR

Q - 6 (a) Explain the one wattmeter method for measurement of power in 3-phase circuits. [07]

(b) List out the methods of solving parallel A.C.circuits. and explain the any one with neat diagram. [07]
